

SOC109 — Emotet Malware Detection & Response Investigation

LetsDefend SOC Analyst Simulation | Role: Tier 1 SOC Analyst | MITRE ATT&CK: T1204 – User Execution

Alert	Final Verdict	Severity	Playbook Score
SOC109 – Emotet Malware Detected	True Positive	Medium	100%

1. Summary

A malicious Word document (**1word.doc**) triggered an Emotet detection on host **RichardPRD**. The endpoint's antivirus reported the file as **Cleaned**, but log analysis showed the document had already spawned a suspicious child process that attempted to download a second-stage executable from an external domain. That domain was confirmed malicious via VirusTotal and urlscan.io. Log Management was then checked for evidence that the endpoint actually reached the malware's C2 infrastructure — it had not. The case was closed as a **True Positive**: malware executed and reached out externally, but was cleaned before a successful C2 connection was established.

2. Alert Details

Field	Value
Rule	SOC109 – Emotet Malware Detected
Severity	Medium
Alert Type	Malware
Difficulty	Medium
Hostname	RichardPRD
Source IP	172.16.17.45
File Name	1word.doc
File Size	188.95 KB
File Hash (MD5)	349d13ca99ab03869548d75b99e5a1d0
Device Action	Cleaned

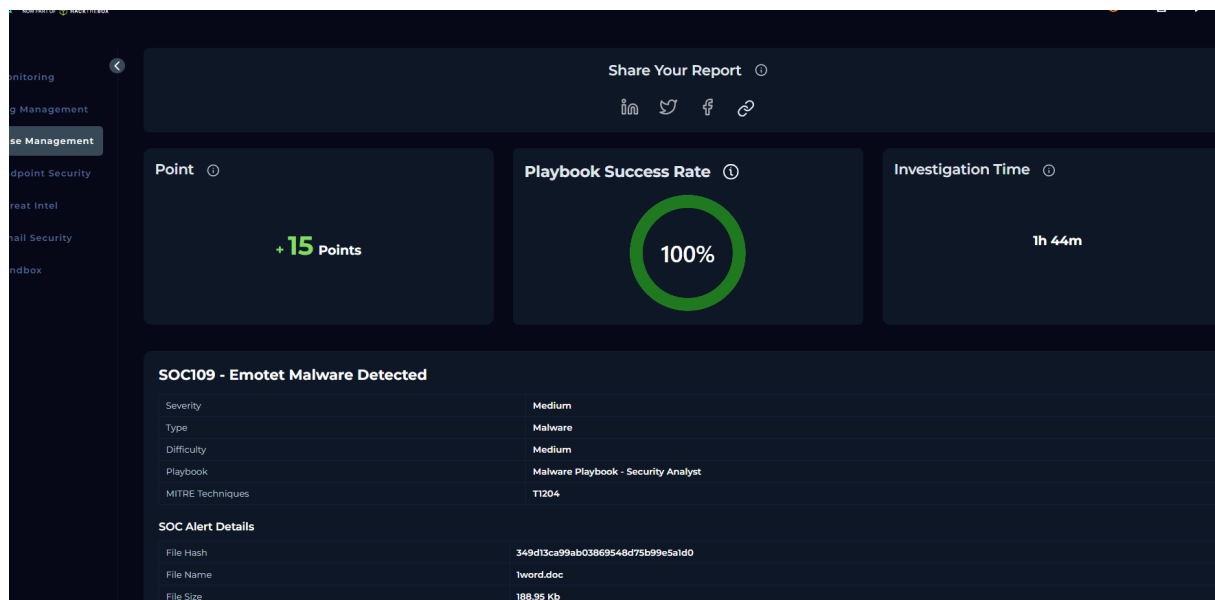


Figure 1 — Case summary: points, playbook success rate, and alert details.

A .doc file spawning outbound executable requests is a textbook Emotet/macro-dropper pattern — the file itself doesn't need to be flagged as malicious for the behaviour it triggers to be.

3. Investigation Steps

Step 1 — Identify the malicious network activity

Searched Log Management using the source IP (172.16.17.45) and found a proxy event showing an outbound request for an executable.

Field	Value
Source Address	172.16.17.45
Destination Address	5.135.143.133
Destination Port	443
Process	EQNEDT32.EXE
Parent Process	excel.exe
Request Method	GET
Requested URL	http://andaluciabeach.net/image/network.exe

The process chain (Office → EQNEDT32.EXE) is a known exploit pattern for malicious Office documents delivering a payload without user awareness.

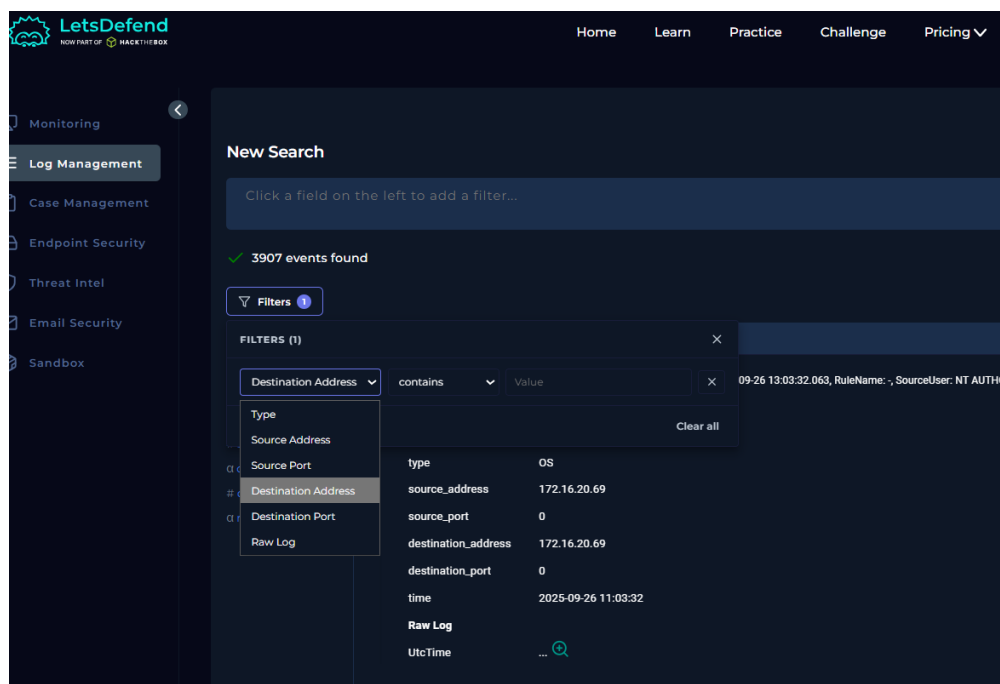


Figure 2 — Searching Log Management for outbound activity from the affected host.

Step 2 — Analyze the malicious URL

Ran the URL through third-party analysis tooling (VirusTotal, urlscan.io) rather than assuming intent from the log alone.

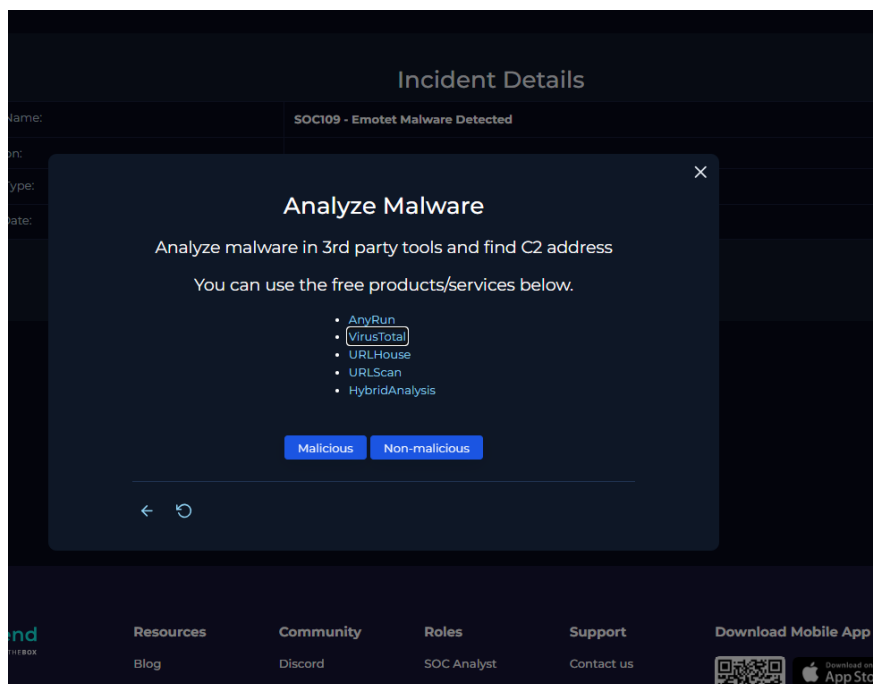


Figure 3 — Selecting VirusTotal to analyze the malicious URL.

VirusTotal flagged the URL **9/92**, with multiple vendors classifying it as Malware or Phishing.

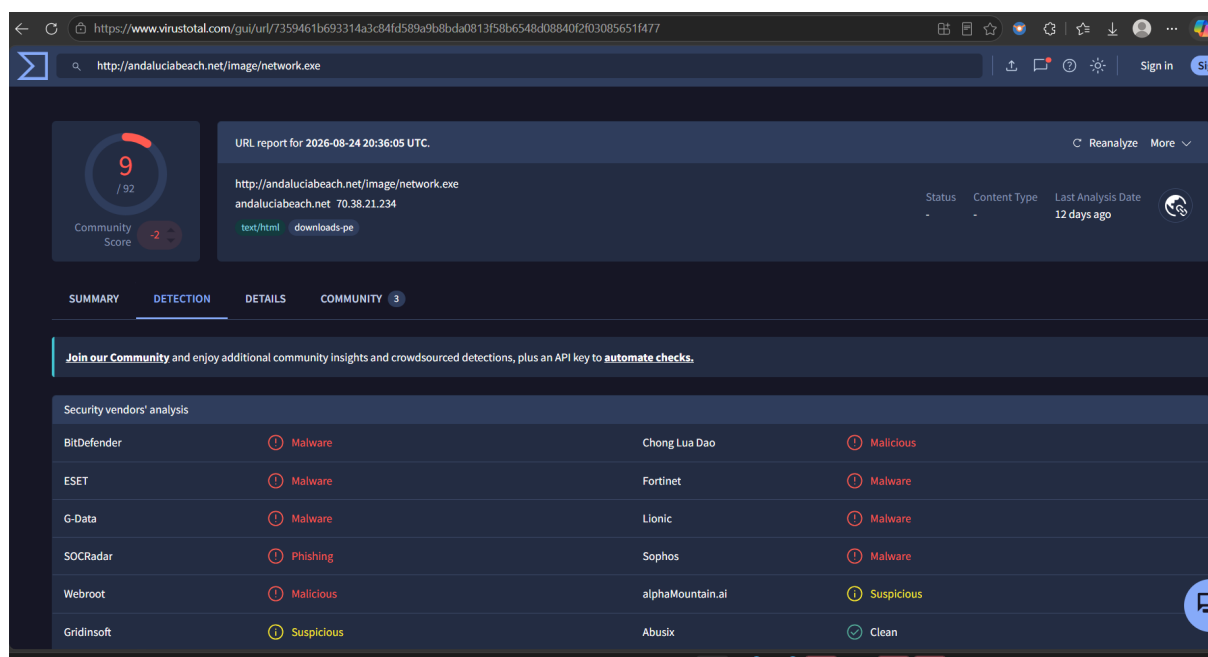


Figure 4 — VirusTotal detection results for the malicious URL.

urlscan.io independently confirmed the domain as malicious and tagged it under “Malware Sites” / “spyware and malware.”

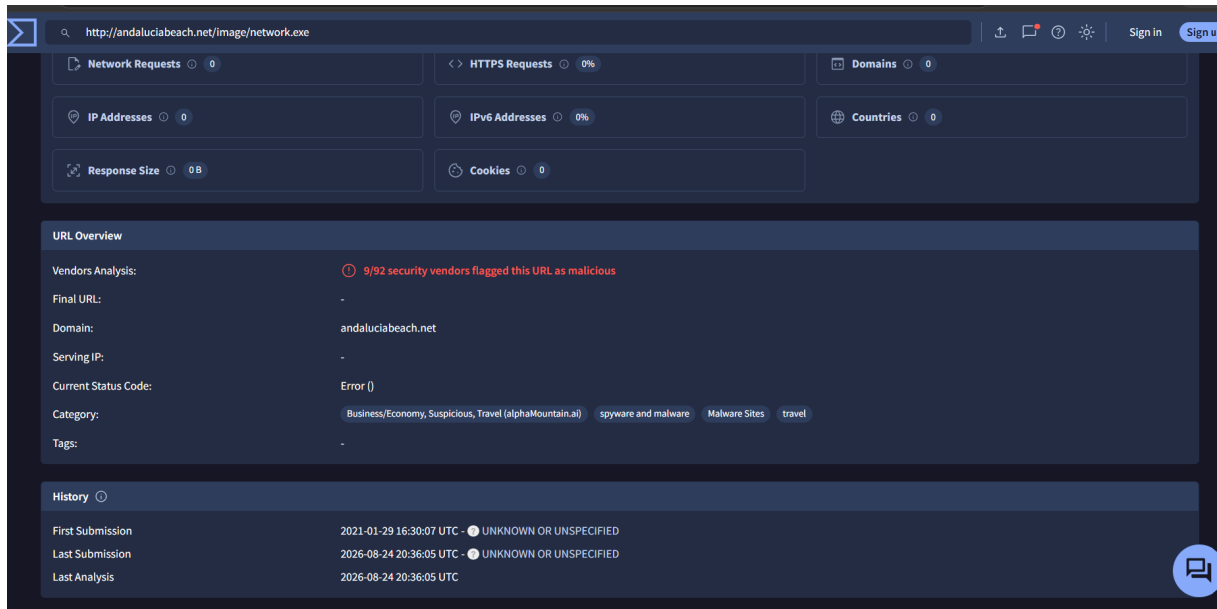


Figure 5 — urlscan.io overview confirming the malicious classification.

Two independent sources agreeing on the same verdict is stronger evidence than relying on a single tool.

Step 3 — Check whether the endpoint reached the C2

Confirming malware exists is not the same as confirming impact. Searched Log Management for any traffic from RichardPRD to the identified malicious infrastructure to determine whether the connection actually succeeded.

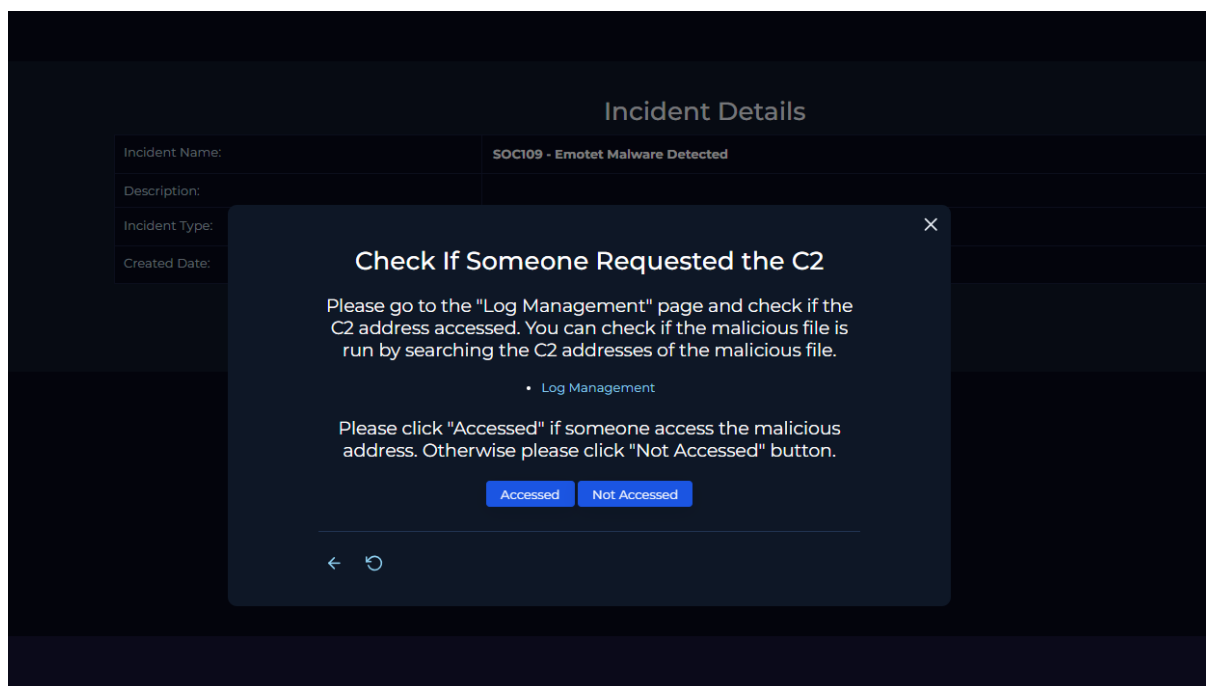


Figure 6 — Checking Log Management for evidence the C2 address was accessed.

No successful outbound connection to the C2 infrastructure was found — the download request was made, but there was no confirmed follow-through traffic, and the endpoint's antivirus had already cleaned the payload.

Step 4 — Close the case

With malware confirmed, malicious infrastructure identified, and no evidence of successful C2 communication, the incident was submitted as a **True Positive**.

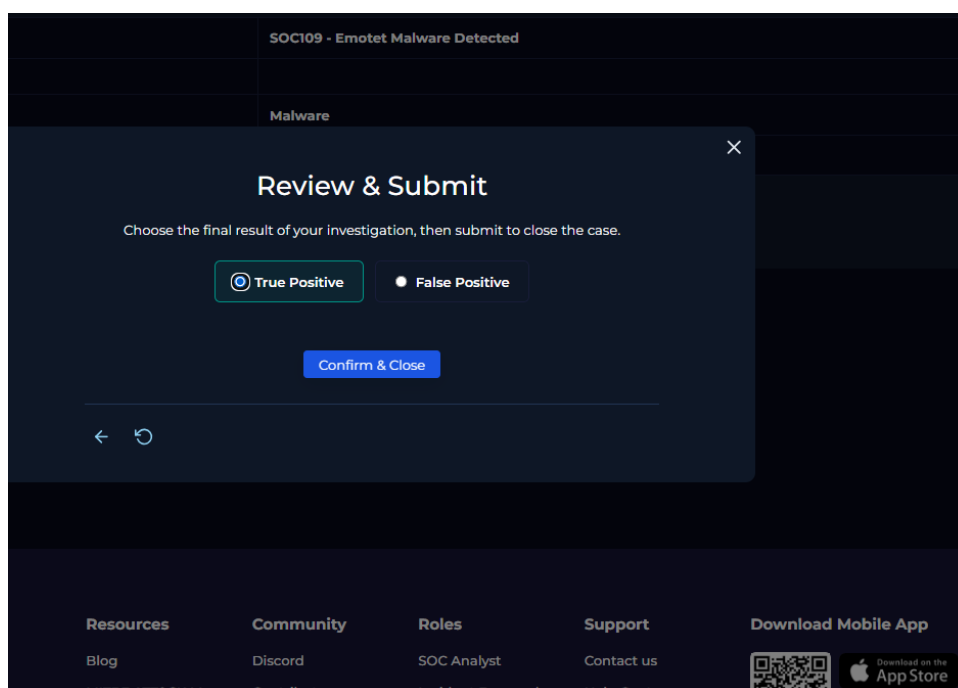


Figure 7 — Case submitted as True Positive.

4. Investigation Performance Summary

Metric	Result
Points Earned	+15
Playbook Success Rate	100%
Investigation Time	1h 44m

5. Indicators of Compromise (IOCs)

Type	Indicator
File Name	1word.doc
MD5 Hash	349d13ca99ab03869548d75b99e5a1d0
File Size	188.95 KB
Internal Host	172.16.17.45 (RichardPRD)
Malicious Domain	andaluciabeach.net
Malicious URL	http://andaluciabeach.net/image/network.exe
Suspicious Destination IP	5.135.143.133:443

6. MITRE ATT&CK; Mapping

Tactic	Technique	Why it applies
Execution	T1204 – User Execution	The infection chain relied on a user opening a malicious Word document, which

7. Findings & Recommendation

- **Verdict:** True Positive — malicious document executed and attempted to reach external infrastructure; antivirus cleaned the payload before a confirmed C2 connection.
- **Recommended remediation:** block andaluciabeach.net and 5.135.143.133 at the perimeter, verify the AV cleaning action fully removed all artifacts on RichardPRD, and search the environment for the same file hash or domain on other hosts.
- **Lesson learned:** this case shows why C2 confirmation matters — assuming impact from a detected file alone would have overstated the severity; checking logs for actual outbound success gave an accurate, defensible verdict.

8. Skills Demonstrated

SIEM alert triage | Log analysis | Malicious document (macro/dropper) pattern recognition | OSINT threat intelligence (VirusTotal, urlscan.io) | IOC extraction | MITRE ATT&CK; mapping | Incident documentation

Disclaimer: This investigation was carried out in a simulated SOC environment (LetsDefend) for training purposes. All hosts, file hashes, and IP/domain indicators belong to the training scenario and are not from a real organisation's environment.